School of Electronic Engineering and Computer Science

EBU6304 – Software Engineering Group Project

35% coursework.

International School Teaching Assistant Recruitment System

- Software Development Using Agile Methods

1. General Information

In the coming weeks, your team will develop a software application that will be used by BUPT International School for recruiting Teaching Assistants. Agile methods should be applied throughout all stages of development, including requirements, analysis and design, implementation, and testing. Iterations (Sprints) should be carefully planned and documented to demonstrate incremental delivery, continuous feedback, and reflective improvement.

There are no restrictions on what may be included in the final product, and the specification provided in this handout contains only high-level, abstract requirements. It is important to recognise that determining software requirements is one of the most critical and complex phases of any development project. You should apply appropriate requirement-finding techniques and Agile practices to identify the actual requirements at a suitable level of detail. Most importantly, you must prioritise the features to be implemented through iterations, considering both feasibility and customer needs. As in real-world software development, you should define the project scope clearly and appropriately. Keep your design simple, modular, and extensible. Justifying your decisions, process, trade-offs, and design choices is an essential part of the coursework.

Key Dates:
Handout release date: **6th March 2026**
First assessment: **22nd March 2026**
Intermediate assessment: **12th April 2026**
Final assessment: **24th May 2026**

电子工程与计算机科学学院

EBU6304——软件工程组项目

35%的课程作业。

国际学校助教招聘制度研究

- 敏捷方法在软件开发中的应用

1. 一般信息

未来数周内，贵团队将开发一款供 BUPT 国际学校招聘教学助理使用的软件应用。敏捷方法应贯穿开发全流程，涵盖需求分析、设计、实施及测试等阶段。需精心规划并记录迭代（Sprint）过程，以实现增量交付、持续反馈和反思性改进。

最终产品的内容不受限制，本手册提供的规范仅包含高层次的抽象要求。需要特别注意的是，确定软件需求是任何开发项目中最关键且复杂的阶段之一。应运用适当的需求发现技术与敏捷实践，在合理细节层级识别实际需求。最重要的是，必须通过迭代优先级排序待实现功能，兼顾可行性与客户需求。如同真实软件开发，需清晰界定项目范围。保持设计简洁、模块化且可扩展。对决策依据、流程安排、权衡考量及设计选择进行论证，是课程作业的重要组成部分。

关键日期：
手册发布日期：**2026年3月6日**
第一次评估：**2026年3月22日第22次**
中期评估：**2026年4月12日**
最终评估日期：**2026年5月24日（星期二）**

2. Project Specification

2.1 Basic Requirements

BUPT International School recruits Teaching Assistants (TA) each semester to support academic modules and various school activities. Currently, the application process relies on forms and Excel files. A software application is needed to streamline the workflow. You will act as an Agile software development team to develop a simple recruitment system.

The system should address the specific needs of BUPT International School to recruit TAs for Module Organisers (MO) and other activities (such as invigilation). While part of the task requires you to analyse the recruitment process and identify requirements using suitable techniques, here are some suggested functions to assist you in getting started. Suggested features can include the following, but not limited to:

- TA can create an applicant profile
- TA can upload CV
- TA can find available jobs
- TA can apply for jobs
- TA can check application status
- MO can post jobs
- MO can select applicants for jobs
- Admin can check TA's overall workload

Some AI-powered features could include:

- Matching skills between jobs and applicants
- Identifying missing skills for applicants
- Balancing workload

A complete prototype of the application must be produced. While it is not required to implement fully functional code for every feature represented in the prototype, the team must implement a selected set of CORE functions. These core features should demonstrate the system's primary value.

2.2 Other Requirements (mandatory)

- The software must be developed as either
  a) A stand-alone Java application, OR
  b) A lightweight Java Servlet/JSP Web-based application.
- All input and output data should be stored in simple text file formats. For example plain text (.txt), CSV, JSON, or XML. Do not use a database.

**Adhering to the above restrictions is essential to ensure 1: focus remains on fundamental Software Engineering principles, rather than on the complexities of frameworks or languages. Introducing advanced tools, such as Spring Boot or database integration, could divert attention toward framework-specific features and configuration, which are not the primary learning objectives of this module. 2: the system remains compatible with existing data and interoperable with legacy systems. 3: fairness across all students with the necessary prerequisite knowledge and skills.*

2. 项目规范

2.1 基本要求

BUPT 国际学校每学期都会招募教学助理（TA）以支持学术课程和各类学校活动。目前，申请流程依赖于表格和Excel文件。需要一款软件应用来简化工作流程。您将作为敏捷软件开发团队，开发一个简单的招聘系统。

该系统需满足 BUPT 国际学校为模块组织者（Mo）及其他活动（如监考）招募教学助理的具体需求。虽然部分任务要求您运用适当方法分析招聘流程并确定需求，但以下功能建议可助您快速上手。建议功能包括但不限于：

- TA可创建申请人档案
- TA可上传简历
- TA 可找到可用工作
- TA可以申请工作
- TA可检查应用程序状态
- 摩可以发布工作
- Mo可筛选职位候选人
- 管理员可查看TA的总体工作量

一些人工智能功能可能包括：

- 求职者与职位的匹配度
- 识别申请者所欠缺的技能
- 工作量平衡

必须制作应用程序的完整原型。虽然不需要为原型中所代表的每个功能实现完全功能的代码，但团队必须实现一组选定的核心功能。这些核心功能应展示系统的首要价值。

2.2 其他要求（强制性）

- 该软件必须开发为以下任一形式
  a) A独立的Java应用程序，OR
  b) 一个轻量级的Java Servlet/JSP Web应用程序。
- 所有输入和输出数据应存储在简单的文本文件格式中。例如纯文本(.txt)、CSV、 JSON 或XML。不要使用数据库。

**遵循上述限制至关重要，以确保：1. 重点始终聚焦于软件工程的基本原则，而非框架或编程语言的复杂性。引入Spring Boot或数据库集成等高级工具，可能会使注意力转向框架特定功能和配置，而这并非本模块的主要学习目标。2. 系统需与现有数据保持兼容，并能与遗留系统实现互操作。3. 对所有具备必要先决知识和技能的学生保持公平性。*

Your tasks include defining detailed requirements, designing, developing, and testing the software described above using Agile methodologies. You have the freedom to design the system as you see fit, provided it meets the core customer needs and the project scope is clearly defined.

2.3 AI-Assisted Development

You are encouraged to use AI to assist various aspects of the project, including brainstorming, story writing, prototyping, design, coding, debugging and testing. However, you should not rely solely on AI-generated content, as AI tools have limitations. Outputs may lack domain-specific knowledge, contain subtle logic errors, produce inefficient code, or overlook ethical and privacy considerations. Additionally, AI solutions are often generic and may not reflect the critical thinking and adaptability required for real-world problem-solving. To ensure understanding and avoid over-reliance, you should be prepared to answer questions during live demonstration, such as "Why did AI make this recommendation?" or "How did you modify AI's solution?" to demonstrate your reasoning and modifications.

If you implement AI-powered functions (e.g. skill matching, recommendation), you must integrate AI responsibly and critically. AI-generated outputs must not be blindly accepted. Your system must combine structured logic with AI-based interpretation and provide explainable results.

3. Agile Management and Version Control

Each group has 6 students. You are the Agile team working together to complete the project. All students in a group must work on ALL aspects of the project, to obtain full software engineering skills. You should use the techniques you have learnt in the lectures to manage the project, e.g., Scrum, daily standup meetings, working around a table, scrum master and decision making, etc.

Each team member must have a visible branch and corresponding commits on GitHub for their assigned development tasks. **These will serve as key evidence of individual contributions**.

Each group will be assigned a teaching assistant to support, give feedback, monitor the group's progress and individual contribution. Your teaching assistant should be your first point of contact for questions or issues. The teaching assistants will regularly check both your group's progress and individual contributions. Teaching assistants are NOT responsible for marking the project. Marking is done by the lecturers in the Software Engineering teaching team. Make sure your teaching assistant has access to your GitHub repository. Refer to the teaching assistant allocation file on QMPlus to see the contact of your allocated teaching assistant.

All members of group should be collaborators to the main project repository. The main project repository (i.e. the master branch) MUST be accessible to the assigned teaching assistant. The teaching assistant may request to access the sub-branches if necessary.

你的任务包括定义详细的需求，使用敏捷方法设计、开发和测试上述软件。你可以自由地设计系统，只要它满足核心客户需求，并且项目范围明确。

2.3 人工智能辅助开发

我们鼓励您运用人工智能技术协助项目各环节，包括头脑风暴、故事创作、原型设计、编程开发、调试测试等。但需注意，切勿完全依赖AI生成内容，因为AI工具存在局限性。其输出可能缺乏领域专业知识、存在细微逻辑错误、生成效率较低的代码，或忽视伦理与隐私考量。此外，AI解决方案往往缺乏针对性，可能无法体现解决实际问题所需的批判性思维与适应能力。为确保理解并避免过度依赖，建议您在实时演示时主动回答“为何采用该方案？”或“如何优化AI解决方案？”等提问，以展示您的分析思路与改进方案。

若需实现人工智能驱动功能（如技能匹配、推荐系统），必须以审慎且批判性的方式整合AI技术。AI生成的输出结果不可盲目采纳，系统需将结构化逻辑与AI解析相结合，并提供可解释的输出结果。

3. 敏捷管理与版本控制

每个小组由6名学生组成，你们将组成敏捷团队共同完成项目。小组所有成员必须参与项目的全部工作，以全面掌握软件工程技能。你们需要运用课堂所学的项目管理技术，例如Scrum（敏捷开发）、每日站会、围桌协作、Scrum主管角色及决策机制等。

每个团队成员必须在GitHub上为他们分配的开发任务有一个可见的分支和相应的提交。**这些将作为个人贡献的关键证据**。

每个小组将配备一名教学助理，负责提供指导、反馈、监督小组进度及个人贡献。遇到问题或疑问时，请优先联系你的教学助理。教学助理会定期检查小组整体进度和个人贡献情况，但不负责项目评分工作，评分由软件工程教学团队的讲师完成。请确保教学助理能访问你的GitHub仓库。具体联系方式请查阅QMPlus平台上的教学助理分配文件。

所有小组成员均应作为主项目仓库的协作者。主项目仓库（即主分支）必须对指定助教开放访问权限。必要时，助教可申请访问子分支。

GitHub should be used explicitly to demonstrate your contributions, through activities such as through commits, forks, pull requests, Issues, and README updates. Each member must actively participate to demonstrate clear engagement and contribution within the team. Insufficient evidence on GitHub will be considered inadequate contribution during marking, with no exceptions.

4. Timeline

These are suggested timeline of activities and deliverable for each week. There are 8 scheduled lab sessions in BUPT teaching weeks 3,4,5,6,9,10,11,12, you may use the scheduled lab time for project delivery and personal reflection.

BUPT week	Scheduled Lab	Activities and Deliverable	Assessment Point	Weight (%)
1	-	Group and GitHub Repository Setup	-	-
2	-	Story writing workshop Prototype with user feedback	-	-
3	22 March	Iteration 1.	First assessment • Online submission on QM+ 1. Product Backlog 2. Prototype 3. A brief report • GitHub	30%
4	29 March	Iteration 1 continues. Working Software version 1.	-	-
5	5 April	Iteration 2	-	-
6	12 April	Iteration 2 continues. Working Software version 2.	Intermediate assessment • Demonstration and Viva • GitHub	20%
7	-	Iteration 3.	-	-
8	-	Iteration 3 continues. Working Software version 3.	-	-
9	3 May	Iteration 4	-	-
10	10 May	Iteration 4 continues. Working Software version 4.	-	-
11	17 May	Preparing for final delivery.	-	-
12	24 May	Final Product Delivery	Final assessment • QM+ submission: 1. Software 2. Demonstration video 3. A short report • Demonstration and viva • GitHub	50%
				100%

应通过提交代码、创建分支、发起拉取请求、提交问题及更新readme文档等方式，明确展示个人贡献。每位成员需积极参与，以证明其在团队中的明确参与度和贡献。若在GitHub上提交的证据不足，将被视为未达标贡献，此规定不作例外。

4. 时间轴

以下是每周活动与成果的建议时间安排。 BUPT 教学周3、4、5、6、9、10、11、12共安排8次实验课，您可利用这些实验课时间完成项目交付和个人反思。

北京邮电大学 一星期	预定 实验室	活动与交付成果	评估点	权重 (%)
1	-	群组与GitHub仓库设置	-	-
2	-	故事创作工作坊原型与用户反馈	-	-
3	22 行军	迭代1。	首次评估 • QM+ 1. 产品待办列表的在线提交 2. 原型 3. 简要报告 • GitHub	30%
4	29 行军	迭代1继续。 工作软件版本1。	-	-
5	5 四月	第2次迭代	-	-
6	12 四月	迭代2继续。 工作软件版本2。	中期评估 • 演示与Viva • GitHub	20%
7	-	迭代3。	-	-
8	-	迭代3继续。 工作软件版本3。	-	-
9	3 可以	第4次迭代	-	-
10	10 可以	第4次迭代继续。 工作软件版本4。	-	-
11	17 可以	准备最后交付。	-	-
12	24 可以	最终产品交付	最终评估 • QM+提交： 1. 软件 2. 演示视频 3. 简短报告 • 演示和活力 • GitHub	50%
				100%

5. Staged Assessment

This coursework will be managed and assessed through three assessment points.

5.1. First assessment – 30%

This involves the product backlog, prototype and a brief report.

QMPlus Submission:

- **The product backlog**, an Excel file (a template is provided on QM+).
  Filename: *ProductBacklog_groupXXX.xlsx*, where XXX is your group number.
  It should contain all user stories with acceptance criteria, priority, estimation, and iteration planning.
- **The prototype**, a PDF file.
  Filename: *Prototype_groupXXX.pdf*, where XXX is your group number.
  It should contain the completed prototype. Only a low-fidelity or medium-fidelity prototype is required.
- **The brief report**, a PDF file (a template is provided on QM+).
  Filename: *Report_groupXXX.pdf*, where XXX is your group number.
  This document should describe the fact-finding techniques, iteration planning, prioritisation, and estimation methods used in the project. (maximum 5 pages).
  It should also include supporting materials (not included in the 5-page limit), such as interview questions, survey results, and customer feedback (if applicable).

Make a .ZIP file contains all above files and submit on QM+. Only the group leader should submit the files on behalf of the whole group.

GitHub: Your GitHub repository will be checked. Make sure it has been set up properly and clear shows members commits.

5.2. Intermediate assessment – 20%

This involves the checking intermediate versions of the software.

Demo & Viva: *This simulates your meeting with the client in real-life scenario where you inform the clients of your progress.* All group members need to attend a demonstration and viva session. You will need to present the progress and demonstrate the latest version of the software (ideally with each member speaking for 2 minutes). Each member will be asked 1 or 2 questions during the viva.

GitHub: Your GitHub repository will be checked for the up-to-date versions of the software and each member's contribution. Make sure all sub-branch contributions are all merged to the master branch by the checking date.

5. 分期评估

本课程将通过三个评估点进行管理和评估。

5.1. 首次评估 – 30%

这包括产品待办事项列表、原型设计及简要报告。

QMPlus提交：

- **产品待办事项**，一个Excel文件（QM+上提供了一个模板）。
  文件名：*ProductBacklog_groupXXX.xlsx*，其中XXX是您的组号。
  该文档应包含所有用户故事，包括验收标准、优先级、估算及迭代计划。
- **原型**，PDF文件。
  文件名：*Prototype_groupXXX.pdf*，其中XXX是您的组号。
  应包含已完成的原型。仅需低保真或中保真原型。
- **简报**，PDF文件（QM+上提供模板）。
  文件名：*Report_groupXXX.pdf*，其中XXX是您的组号。

  本文件应描述项目中使用的事实调查技术、迭代计划、优先级排序和估算方法。（最多5页）。
  还应包括支持性材料（未包含在5页限制内），例如访谈问题、调查结果及客户反馈（如适用）。

制作一个包含上述所有文件的.ZIP文件，并通过QM+提交。仅小组负责人有权代表全体成员提交文件。

GitHub:你的 GitHub 仓库将被检查。请确保已正确设置并清晰显示成员提交记录。

5.2. 中期评估——20%

这涉及对软件中间版本的检查。

演示与问答环节：*该环节模拟您在真实场景中与客户会面的情况，向客户汇报项目进展。*所有小组成员需参加演示和问答环节。您需要汇报项目进展并演示软件最新版本（最好每位成员发言2分钟）。问答环节中每位成员将被问及1-2个问题。

GitHub：将检查您的GitHub仓库中软件的最新版本及每位成员的贡献情况。请确保所有子分支的贡献在检查日期前已全部合并到主分支。

5.3. Final assessment – 50%

This involves the checking the final software, demonstration and report.

QMPlus Submission:

- **The short report,** a PDF file (a template is provided on QM+).
  Filename: *Report_groupXXX.pdf*, where XXX is your group number.
  This report documents the software design strategy, build plan, and testing strategies and techniques, the use of Gen AI etc. It should contain the sections of Group report (maximum 10 pages) and individual statements (no more than 300 words each). More details can be found in the template.

- **The software,** a ZIP file.
  Filename: *Software_groupXXX.zip*, where XXX is your group number.
  It should include the following:
  - Source code.
  - Test programs.
  - Code documentation, such as JavaDocs.
  - A user manual including key screenshots of the application (at least one for each main frame)
  - A readme file, providing instructions for setting up, configuring, and running the software.

- **The demonstration video,** record a short video to demonstrate your software and submit it to QM+. Requirements of the demonstration video:
  - Recording can be made using any screen capture software.
  - No more than 10 minutes.
  - You should demonstrate the running of your software, as if you do the demonstration to sell your product. You should demonstrate the operation of all functions. It's recommended to demonstrate error handling as well.
  - You must explain your operations in the video, your voice must be recorded, and the narration must be in English.
  - The video format does not matter, as you will upload it to QM+ Media.
  - The video must not be post-edited, and it must not contain any AI-generated content.

Demo & Viva: *This simulates your final meeting with the client in real-life scenario where you deliver the final product and hope to gain user satisfaction.* All group members need to attend a demonstration and viva session. You will be required to complete acceptance testing tasks to verify that the software meets the specified requirements. Each member will also be asked 1 or 2 questions during the viva.

GitHub: Your GitHub repository will also be checked for the latest versions of the software and each member's contribution. Make sure all sub-branch contributions are all merged to the master branch by the checking date.

5.3. 最终评估 - 50%

这包括对最终软件、演示及报告的检查。

QMPlus提交：

- **简短报告**，PDF文件（QM+上提供模板）。文件名：*Report_groupXXX.pdf*，其中XXX是您的组号。
  本报告记录了软件设计策略、构建计划、测试策略与技术、Gen AI等的使用情况。应包含小组报告（最多10页）和个人陈述（每篇不超过300字）等部分。更多细节可参阅模板。

- **该软件**，一个 ZIP 文件。
  文件名：*Software_groupXXX.zip*，其中XXX是您的组号。
  应包括以下内容：
  - 源代码。
  - 测试程序。
  - 代码文档，例如JavaDocs。

  用户手册，包含应用程序的关键截图（每个主界面至少一张）

  一个readme文件，提供软件的安装、配置和运行说明。
- **演示视频**，录制一段短视频演示您的软件并提交至QM+。演示视频要求：
  - 可使用任何屏幕录制软件进行记录。
  - 不超过10分钟。
  - 演示软件运行时，应像推销产品那样生动演示。所有功能都需完整展示，建议同时演示错误处理机制。
  - 你必须在视频中解释你的操作，你的声音必须被记录，并且叙述必须是英语。
  - 视频格式无关紧要，因为您将直接上传至QM+ Media平台。
  - 视频不得经过后期编辑，且不得包含任何AI生成内容。

演示与现场反馈环节：*该环节模拟您在真实场景中与客户进行最终会议，交付最终产品并期望获得用户满意度。*所有小组成员需参加演示与现场反馈环节。您需要完成验收测试任务以验证软件是否符合指定要求。每位成员在反馈环节中还需回答1-2个问题。

GitHub：你的GitHub仓库也会被检查软件的最新版本和每位成员的贡献。确保所有子分支的贡献在检查日期前都已合并到主分支。

6. Marking Criteria

For each assessment stage, **your marks = group marks × individual factor**.

First assessment:

Group Marks

Component	Detail	Max Marks
Product Backlog	The correctness of defining the users and scope	2
	The correctness of writing user stories	5
	The completeness of product backlog	3
Prototype	Functionality	5
	Usability	5
Report	Describe the fact-finding techniques, iteration plan, prioritisation and estimation methods used in the project.	5
Evidence	Sufficient evidence to support the statements in the report.	5
Total		**30**

Individual Factor

Component	Detail	Factor
Contributions	Regular commits on GitHub	1/2
	Observations from the teaching assistant	1/2
Total		**1**

Intermediate assessment:

Group Marks

Component	Detail	Max Marks
Software Version 1 & 2 (GitHub)	Decent progress has been made for each version. The improvement of each version addresses existing problems.	5
Project Management (GitHub)	The GitHub network shows active engagement from all members with respective branches contributing and merged for the release of the new version.	5
Demo & Viva	Excellent teamwork in presenting the current version of software. All questions are answered clearly.	10
Total		**20**

6. 标记标准

每个评估阶段，**你的分数 = 组分数× 个人因素。首次评估：**

群标记

组件	详情	马克·马克斯
产品积压	用户与范围界定的正确性	2
	用户故事的正确书写	5
	产品待办事项列表的完整性	3
原型	功能	5
	可用性	5
报告	阐述项目中采用的事实调查技术、迭代计划、优先级排序及估算方法。	5
证据	有充分证据支持报告中的陈述。	5
共计		**30**

个体因素

组件	详情	因子
捐款	在 GitHub 上定期提交	1/2
	助教观察	1/2
共计		**1**

中期评估：

群标记

组件	详情	马克·马克斯
软件版本1和2 (GitHub)	各版本均取得显著进展，其改进内容均针对现有问题。	5
项目管理 (GitHub)	GitHub网络显示所有成员均积极参与，各分支均对新版本的发布做出了贡献并完成合并。	5
演示与欢呼	团队在当前版本软件的展示中表现出色，所有问题均得到清晰解答。	10
共计		**20**

Individual Factor

Component	Detail	Factor
Demo & Viva	Presentation	1/2
	Q&A	1/2
Total		**1**

Final assessment:

Group Marks

Component	Detail	Max Marks
Demonstration video and software	Design	10
	Implementation	10
	Testing	5
Short report	Content quality	10
Project Management (GitHub)	The GitHub network shows active engagement from all members with respective branches contributing and merged for the release of the new version.	5
Demo & Viva	Pass all acceptance testing tasks. All questions are answered clearly.	10
Total		**50***

Individual Factor

Component	Detail	Check
Contribution	Contribution on GitHub	1/4
	Reflection in the Report	1/4
Demo & Viva	Testing task	1/4
	Q&A	1/4
Total		**1**

*** A student who demonstrates significant contribution and leadership in the group may receive up to 10% additional marks added to the group's final assessment mark.**

个体因素

组件	详情	因子
演示与欢呼	演示文稿	1/2
	Q&A	1/2
共计		**1**

最终评估：

群标记

组件	详情	马克·马克斯
演示视频和软件	设计	10
	实施	10
	测试	5
简报	内容质量	10
项目管理（GitHub）	GitHub网络显示所有成员均积极参与，各分支均对新版本的发布做出了贡献并完成合并。	5
演示与欢呼	通过所有验收测试任务。所有问题均得到明确解答。	10
共计		**50***

个体因素

组件	详情	勾选
贡献	GitHub贡献	1/4
	报告中的反思	1/4
演示与欢呼	测试任务	1/4
	Q&A	1/4
共计		**1**

*** 在小组中表现突出并发挥领导作用的学生，可获得最高10%的加分，计入小组最终评估成绩。**